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To cite this article: Sara Nocco & Marco Cremaschi (25 Mar 2026): Uncovering the Sound of Immersion: EEG and Self-Reported Correlates of Diegetic Audio in Gaming, International Journal of Human-Computer Interaction, DOI: [10.1080/10447318.2026.2638546](https://doi.org/10.1080/10447318.2026.2638546)

To link to this article: <https://doi.org/10.1080/10447318.2026.2638546>



Published online: 25 Mar 2026.



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Uncovering the Sound of Immersion: EEG and Self-Reported Correlates of Diegetic Audio in Gaming

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ABSTRACT

Immersion and flow are central to the gaming experience, yet their real-time physiological correlates remain elusive. This study examines the effect of sound, specifically diegetic music, on player immersion and flow by combining electroencephalographic (EEG) data from a commercial device (Muse 2) with self-reported questionnaires. Thirty participants played a custom platform-style video game under three sound conditions: consistent diegetic, inconsistent diegetic, and no audio. Post game self-reports measured flow, competence, challenge, and immersion in response to sound. Simultaneously, EEG features were analysed both across the four electrodes and considering their average. Machine learning analysis revealed key channel-feature combinations predictive of high immersion and flow states, whereas statistical tests aimed to identify features and channel-feature combinations that were statistically different across the sound conditions. Our findings suggest that, even with a low resolution EEG setup, it is possible to identify potential markers of immersion and flow that are worth further investigation.

KEYWORDS

Immersion; flow; EEG; diegetic audio; affective gaming

1. Introduction

The concepts of immersion and flow have long been debated in psychology, media studies, and human-computer interaction (Michailidis et al., 2018). One accepted definition of immersion refers to it as the complete involvement with something, or the feeling of being surrounded by it (Nilsson et al., 2016). In the context of games, it is associated with profound absorption in the game experience, to the extent that players may describe themselves as being “in the game.” Flow is described as “the optimal psychological experience that can arise from being engaged in an activity” (Csikszentmihalyi, 1990). According to the same author, it is characterized by “the balance between perceived challenges and perceived skills” and it is built on a theory that qualifies it based on nine major components (Csikszentmihalyi, 1975), which are the following: (i) challenge-skills balance, (ii) action-awareness merging, (iii) clear goals, (iv) unambiguous feedback, (v) concentration on the task at hand, (vi) sense of control, (vii) loss of self-consciousness, (viii) transformation of time, and (ix) autotelic experience. A review (Michailidis et al., 2018) on the relation between immersion and flow argues that these concepts do not substantially differ and, thus, more evidence is needed to justify their separation.

Regardless of the conceptual debate distinguishing the two constructs, understanding the drivers and the corresponding physiological correlates of flow and immersion can be beneficial in various contexts. In the domain of video games, the inclusion of features that induce immersion and flow is core to the user experience. Such features play a crucial role in game enjoyment, engagement, and overall success, and are therefore a central focus in game design and prototyping. We identified eight elements that may be identified as primary drivers of the players’ shift to an immersive mode: (i) storytelling (Birk et al., 2016; Moser & Fang, 2014), (ii) game interface design (User Interface (UI) - Head-Up Display (HUD)) (Ahmad et al., 2017; Malaspina et al., 2023), (iii) sound design (Cassidy & MacDonald, 2009) and engine (Firat et al., 2022), (iv) player’s identity (Kukshinov, 2024), (v) equipment (Banks &

Bowman, 2021), (vi) reality (Rivera & Gutierrez, 2023), (vii) controllers (McGloin et al., 2013; Normoyle & Jörg, 2018), and (viii) utilized platform (Omar & Mohamad Ali, 2011). These elements do not constitute an established framework, but rather reflect a conceptual grouping emerging from the reviewed literature and intended to support the scope and focus of the present work.

Although self-assessment questionnaires, such as the Game Experience Questionnaire (GEQ) (Poels et al., 2007), the Immersive Experience Questionnaire (Jennett et al., 2008), and the Game Immersion Questionnaire (Cheng et al., 2014), have been widely used to assess immersion, these subjective measures are prone to biases, including social desirability and recall bias (Choi & Pak, 2005). Furthermore, essentially constituting retrospective self-reports, using self-assessment questionnaires alone fails to capturing the real-time fluctuations in immersive experiences, which can be crucial for understanding the dynamic interplay between game elements and player engagement.

Conversely, physiological signals (such as heart rate variability, electrodermal activity, and facial electromyography) offer promising support to information acquired via questionnaires by providing objective and continuous measures of physiological arousal, which is reportedly associated to flow (van der Linden et al., 2020; Tian et al., 2017), immersion (Hofmann et al., 2021), and emotional states (Juvrud et al., 2022). Previous research has demonstrated the feasibility of using physiological signals, especially electroencephalographic (EEG), to assess engagement and emotional responses in various contexts, including music listening and film viewing (Kim & André, 2008; Li & Chen, 2006). However, the specific applicability of these measures to discriminate different levels of immersion in video games remains largely unexplored. While player experience has previously been investigated using physiological measures such as electrodermal activity (EDA) and electromyography (EMG) (Nacke & Lindley, 2010; Nacke et al., 2010), EEG-based approaches have been less frequently adopted. For instance (Kannegieser & Ratz, 2021; Nacke et al., 2011), employed EEG to assess flow and immersion during gameplay, yet, without a specific focus on the role of audio and using research-grade EEG systems rather than commercial devices. Other studies have explored related constructs using low-density EEG systems: (Plotnikov et al., 2012) investigated flow versus boredom, and (Abujelala et al., 2016) examined enjoyment in video games, both using four-electrode setups; however, neither study focused on audio effects nor discussed the feasibility of employing such devices. Conversely (Garner, 2013), analyzed the impact of audio in video games using EEG, but targeted fear-related responses rather than immersion or flow, and relied on a 14-electrode system while considering only basic statistical features.

Considering this complex framework, the present study aims to preliminarily assess the characterization, in terms of features and spatial localization, of the neural responses associated to flow and immersion in video games, with a particular focus on sound as a sensory driver of these conditions. In fact, given that both constructs are influenced by the multiplicity of game elements listed above, simultaneously addressing all of them would confound the identification of their correlated neurophysiological underpinnings. Therefore, the idea is to adopt a modular approach, isolating in this case sound design, and particularly the use of diegetic sound, as the experimental variable, to explore its specific contribution to immersive and flow states in gameplay. Diegetic sound refers to audio elements that originate from within the game world and that are perceivable by both the player and the in-game characters (e.g., music playing on an in-world device, environmental ambience, character actions). In contrast, non-diegetic or extradiegetic sound consists of audio elements external to the game world, such as background music or UI cues, which the player can hear but the game characters cannot. Although non-diegetic music is more prevalent in commercial games, diegetic audio plays a key role in shaping narrative context, spatial coherence, and the sense of presence, and in general has been linked to higher immersion and fun (Grimshaw et al., 2008; Nacke et al., 2010). Notably, as explained above, its contribution to psychological constructs such as immersion and flow has received limited empirical attention (Haehn et al., 2023), particularly from a physiological standpoint. Given this gap and the theoretical importance of diegetic cues for player engagement, we decided to focus on diegetic sound as a meaningful target for this investigation. In Section 2.1, we explain how diegetic audio, in the present game, consisted primarily of sound effects originating from the player character's actions (e.g., jumping, falling, collecting items). Throughout the manuscript, the term diegetic audio refers to these in-world

action-related sounds. Considering that we employ an ecological, wearable EEG device, a further aim of this study is to examine the feasibility of using such a device for the proposed task. Particularly, the concern is in exploring whether such a commercial system, despite being characterized by limited spatial resolution and electrode coverage compared to medical-grade devices, may be sufficient in identifying features predictive of immersive or flow conditions. In turn, this would enable potential applications for online, user-adaptive systems, which are capable of modifying game elements in real time to steer the player toward specific psychological conditions.

The proposed investigation unfolds along two primary interests. First, the presence of a physiological difference among video game sessions characterized by a different use of diegetic music is assessed. This is accomplished by considering: (i) features averaged across all the signal acquisition channels, to identify those showing a marked discriminative power at a global level; (ii) specific channel-feature pairs, to identify the most informative combinations of spatial locations and signal descriptors that are both sensitive and detectable via a low-density, consumer-grade EEG system. Second, assuming that the neural correlates serve as objective indicators of the two constructs under study, in the sense that they reflect each subject's true and unmasked physiological response to the experimental conditions, the reliability of self-reported questionnaires in capturing these states is indirectly explored. This is done by attempting to map significant changes among different sound conditions in EEG features to the most discriminative features of high/low self-reported flow and immersion.

The remainder of this paper is structured as follows. [Section 2](#) details the experimental setup, including the game design, audio conditions and underlying assumption, questionnaires, EEG device, experimental procedure, and participant information. [Section 3](#) discusses the methodology adopted to address the research interests, as well as the EEG signal pre-processing and feature extraction procedures. [Section 4](#) and [Section 5](#) present, respectively, results and relative discussion of the experimental analysis, including a comparison with relevant state of the art studies. Finally, [Section 6](#) summarizes the contributions brought by this work, anticipating some planned future developments.

2. Materials

Participants had to play a game specifically developed for this study. Each participant experienced three distinct sound conditions, namely:

1. Extradiegetic sound and consistent diegetic sound turned on;
2. Extradiegetic sound and inconsistent diegetic sound turned on;
3. Sound completely turned off.

These three conditions were conceived with the purpose of inducing different degrees of flow and immersion in players. We expected the first condition to be associated with the highest levels of flow and immersion, and the third condition to yield the lowest levels of these states. The second condition was included with the hypothesis that inconsistent audio might lead to more variable and generally lower levels of flow and immersion compared to the consistent audio condition, by either bothering or amusing players, or triggering attentional mechanisms. What we intend by “inconsistent audio” in this paper is explained in [Subsection 2.1](#). For each condition, questionnaires and physiological responses were recorded. This section details the materials (game, questionnaires, and physiological signal recording device) and the experimental procedure employed in this study.

2.1. Experimental setting

2.1.1. Video game

The custom game was developed using the open-source game engine *Godot*,¹ and its design was inspired by the classic platformer *Super Mario Bros.*² Players could choose to control the game using either the WASD keys or the arrow keys. This input configuration of game commands was designed with future developments in mind, where the non-dominant hand could remain free to allow the placement of a device for acquiring additional physiological signals, in the perspective of a multimodal



Figure 1. Starting screen of the developed game.

approach to immersion detection and affect recognition. An example of such a device is the Shimmer 3,³ which can measure photoplethysmographic and galvanic skin response signals.

Two game versions were developed to implement the different diegetic audio conditions. In the consistent diegetic sound version, the sound of each in-game action is coherent with the players' expectations based on game genre norms. Conversely, in the inconsistent diegetic sound version, the game sounds of the consistent version were paired with a mismatched in-game action throughout the entire gameplay. For instance, the sound associated with collecting coins in the consistent sound version (which is very similar to the one present in *Super Mario Bros.*) was replaced with the jumping sound. This mismatched pairing was maintained constant across each game session and for all the players, i.e., each action was one-to-one reassigned a new sound among those already present in the consistent version.

Unlike most studies in the state of the art, which have relied on fast-paced first-person shooter (FPS) games (Drachen et al., 2010; Grimshaw et al., 2008; Nacke & Lindley, 2010), the presented game features a more linear design, with minimal in-game visual and event distractions, as well as simple, familiar gameplay dynamics. This allows the experimental focus to remain on the auditory dimension only, controlling for the other elements affecting immersion and flow in gameplay. Future studies may further refine this design by accounting for physiological data analysis of emotionally salient events such as the player's death, as this could elicit marked physiological responses that may partially confound the effects of audio-based manipulations. Figure 1 shows the starting game screen.

2.1.2. Questionnaires

Two different questionnaires were used: a pretest questionnaire, administered once at the beginning of the experiment and collecting socio-demographic information, and a post-game questionnaire, which participants completed after each of the three game sessions to report their game experience.

The post-game questionnaire was partially adapted from the GEQ. Of the original seven GEQ dimensions, two (Competence and Challenge) were retained in their original combination of items, whereas four (Tension/Annoyance, Positive Affect, Negative Affect, and Sensory and Imaginative Immersion) were excluded. The Flow dimension was expanded with two additional items: "To what

extent did your feeling of being inside the game environment outweigh your awareness of the real world?” and “Did you notice events happening around you during the game?” In addition to these three dimensions, a newly defined one, which was called “Audio Immersion,” was included to address the auditory component of the gaming experience. This consisted of the following eight items:

- To what extent were you distracted by external sounds?
- Did you feel the need at any point to stop playing in order to see or hear what was happening around you?
- To what extent have you enjoyed the game audio?
- To what extent have you felt bothered by the game audio?
- To what extent have you found the audio entertaining?
- To what extent did the audio help you feel immersed in the game?
- To what extent did the audio distract you during gameplay?
- To what extent did you adapt to the audio as the game progressed?

All the described dimensions were rated on a discrete scale ranging from 1 to 5 (1 = not at all, 5 = a lot). Three items were included to assess self-reported Valence, Arousal, and Liking using a single-item, 9-point numeric rating scale ranging from 1 (*Not at all*) to 9 (*Very much*). Particularly, participants were asked “Did you feel happy?” to measure Valence, “Did you feel excited?” for evaluating Arousal, and “How much did you like the game overall?” to quantify Liking. Valence and Arousal are the two emotional dimensions constituting Russell’s circumplex model of affect (Russell, 1980). Their inclusion, beyond the analyses presented in what follows, was intended to eventually enable the measurement of overall affect and allow for comparisons in future developments with emotion recognition studies, which frequently rely on these dimensions. The Liking item, often included in such studies, was added to capture participants’ personal enjoyment of the game experience. Valence can also be regarded as a comprehensive metric that encompasses both GEQ’s Positive and Negative Affect dimensions, thus capturing such information while simultaneously reducing participants’ cognitive effort due to the questionnaire length (Gibson & Bowling, 2019) and utilizing a well-established measure in the affective computing domain.

Hence, the post-game questionnaire consisted of 7 dimensions (Challenge, Competence, Audio Immersion, Flow, Valence, Arousal, and Liking) captured by 27 items, instead of the original 38 GEQ questions. The complete post-game questionnaire can be found in Table A1 in Appendix A.

The GEQ was selected as the basis for the post-game questionnaire due to its widespread use in the literature as a standard tool for assessing game experience (Johnson et al., 2014). However, it was readapted to better suit the specific goals of this study. This partial adherence to the original questionnaire structure is justified by prior observations in the literature that the psychometric properties of the GEQ have not yet been fully established, leaving room for modifications (Law et al., 2018).

2.1.3. Electroencephalographic device

To record the EEG signal, the Muse EEG headband⁴ was utilized. Muse 2 (Figure 2) is a wearable brain sensing device that captures electrical activity from the scalp and transmits the data to a connected device via Bluetooth. Along with an additional reference electrode, it has four electrodes placed according to the international 10–20 system and that collect signals from the frontal (AF7, AF8) and temporal (TP9, TP10) areas of the cortex, as shown in Figure 2. The lateral electrodes are particularly relevant for the presented study as the temporo-parietal areas are involved in auditory processing and multisensory integration (Schirmer et al., 2012). Muse is primarily used as a personal neurofeedback device. Still, it has also been utilized for research (Krigolson et al., 2017) in several fields, including cognitive neuroscience, and particularly for the identification of mental and emotional states (Alsayigh & Khidhir, 2024; Bird et al., 2018; Rahman & Nasor, 2024). In the context of affective ludology, it has been previously employed to measure engagement (Abujelala et al., 2016), whereas a different EEG device, still consisting of 4 electrodes, was used to assess flow (Plotnikov et al., 2012). Its four dry electrode sensors are embedded into a flexible headband positioned across the forehead and over the

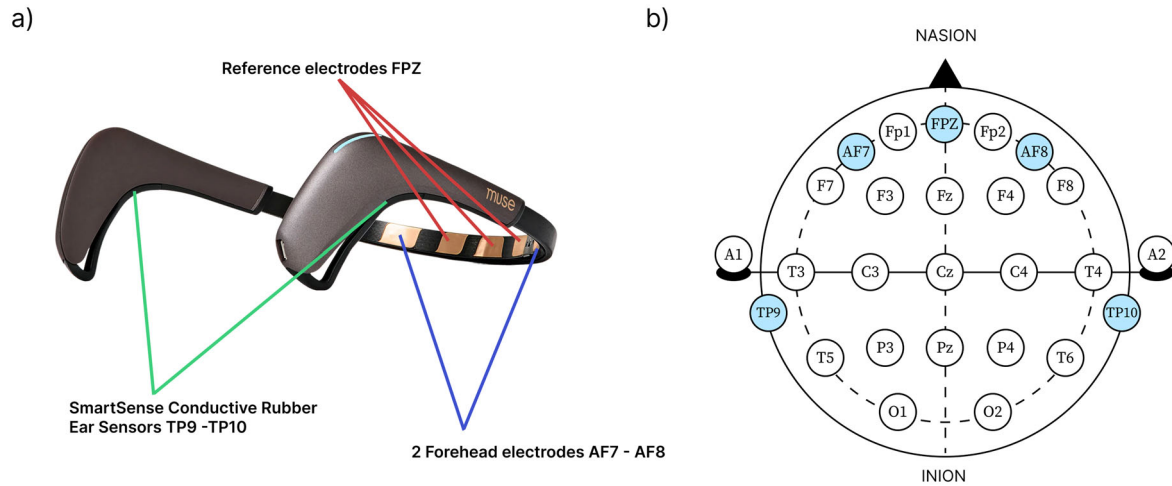


Figure 2. Muse 2 device. (a) Displays the electrodes positioning on the device, while (b) represents the corresponding electrodes location in the international 10–20 system.

mastoid bones behind the ears. With such a compact and lightweight design, Muse was identified as an attractive tool for this research study, to eventually develop a portable online system to detect in real-time the players' perceived levels of flow and immersion and consequently adapt gameplay dynamics.

2.1.4. Subjects

A total of 32 participants (20 males and 12 females), aged between 21 and 41 years ($M = 28.57$, $SD = 5.32$), participated voluntarily in this study with written consent. All participants were recruited within the University of Milano-Bicocca, being either master's students, PhD students, research fellows or professors. During recruitment, they were asked to report any known sensory impairments, particularly auditory impairments, given the nature of the sound-based experimental manipulation. No participant reported hearing-related conditions affecting task performance.

2.2. Experimental protocol

Upon the beginning of the experiment, each participant was asked to adjust the gaming station to guarantee a comfortable experience during gameplay. Verbal instructions were then provided, outlining the available game controls and the objective of the game, i.e., to complete the session while collecting all the coins the player encountered in the virtual environment. The experimental session proceeded as follows. First, participants completed a pretest questionnaire that gathered sociodemographic information. This was followed by a baseline physiological recording lasting two minutes, during which subjects were asked to relax and fixate on a centrally presented cross on the screen. Subsequently, participants engaged in three game sessions, each corresponding to one of the three different sound conditions. The order of delivery of the sound configurations was randomized across participants to mitigate order effects, which could otherwise confound results in within-subjects designs and affect the assumptions underlying repeated measures Analysis of Variance (ANOVA).

Each game session lasted two minutes and featured one of the three sound designs. After each session, participants completed the post-game questionnaire, whose items' order was randomized for each participant across game sessions, followed by a 30-second fixation cross to allow physiological signals to return to baseline. Finally, subjects were asked to comment on their experience during the three sessions. Throughout the experiment, signal quality was checked prior to and continuously monitored during data acquisition to ensure sufficient data quality. Sound volume was kept at a constant and comfortable level for all participants. The sound source was the laptop device that the participant utilized to play.

3. Methodology

This section discusses the methodology adapted to address the research objectives via the collected data, including a subsection (see 3.1) that details EEG signal pre-processing and feature extraction.

Out of the 32 participants involved in data collection, 2 were excluded from the analysis due to issues encountered during the experimental session. One case involved severely bad signal quality on frontal electrodes due to incompatibility between the EEG recording device and the participant's skull shape. The other exclusion was due to a procedural issue that may have significantly affected the participant's cognitive state, potentially introducing bias into the data. As a result, the final analysis was conducted on a sample of 30 subjects.

Except Valence, Arousal, and Liking, whose scores coincide with a single item labeling, the remaining post-game questionnaire component scores (Competence, Challenge, Flow, and Audio Immersion) were computed as the average value of their items, after the direction of some of them had been inverted for consistency of their meaning. For these four components, Cronbach's α was calculated to assess internal consistency reliability.

Following descriptive statistics, a repeated-measures ANOVA was conducted to assess the effect of the different audio conditions across the 30 participants on the 7 post-game questionnaire dimensions. Sound modality was used as the within-subject factor for all the components. Mauchly's test was performed for sphericity, and Greenhouse-Geisser correction was applied when sphericity was violated. To gain further insights on the role of each single audio condition, pairwise paired *t*-tests with Bonferroni correction (Haynes, 2013) for multiple comparisons were performed to understand which specific pairs of conditions differed. We chose one-way repeated measures ANOVA for the way that Competence, Challenge, Flow, and Audio Immersion were computed, resulting in composite measures that are commonly treated as continuous and approximately interval in psychological research. Although they were derived from a single item, the same method was adopted also for Valence, Arousal, and Liking to maintain methodological consistency. As a sensitivity analysis, we additionally implemented non-parametric Friedman tests.

The same methodology (i.e., repeated-measures ANOVA and pairwise paired *t*-tests) was adapted to verify statistically significant differences at physiological level across game sessions, leveraging the extracted EEG features. Particularly, these analyses were run twice, to capture a varying level of detail:

1. retaining for each type of feature its channel location information;
2. averaging the features across the four acquisition channels, producing therefore broader information at the feature level.

Finally, an XGBoost binary classification model was trained to distinguish between high and low values of flow and audio immersion and obtain features importance scores that could provide indications for brain areas to be further investigated. Scores greater than or equal to 3.5 were labeled as high, low otherwise. The threshold was set based on an exploratory analysis aimed at finding a value that could balance the instances in two groups and be common to both dimensions, without excessively adapting to our sample of participants but remaining meaningful for generalization purposes. Features importance was based on the gain score. Furthermore, features having a gain score higher than or equal to the mean gain score for each of the two dimensions were retained, and these two subsets were leveraged to verify whether model performance could benefit from data dimensionality reduction. Leave-one-subject-out (LOSO) was used as a splitting validation strategy, to allow the model to be independent of the individual subject responses and generalize to unseen subjects, therefore giving a subject-independent strategy to the study. Accuracy, precision and F1-score were employed as evaluation metrics.

3.1. EEG signal analysis

Data preprocessing was executed using MATLAB⁵ and the EEGLAB toolbox.⁶ EEG signals were not downsampled, maintaining the original sampling frequency of 256 Hz to preserve the entirety of the

provided information. The signals were bandpass filtered using a FIR filter in the range of 1–60 Hz to retain the δ (1–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and lower γ rhythms (30–60 Hz). Moreover, a notch filter (50 Hz for Europe) was used to remove the power line noise. Following the filtering step, segmentation was performed to divide each signal into the different experimental conditions, as defined by the experimental protocol. Finally, for each subject and for each acquisition channel, EEG signals were standardized in amplitude according to a Z-score standardization to reduce inter- and intra-subject variability.

Two groups of features were extracted: Power Spectral Density (PSD) and Hjorth parameters. PSD is a frequency domain feature that describes the distribution of power in a signal with respect to its frequency components, therefore providing a way to analyze the signal's frequency content. For this analysis, it is estimated on five different frequency ranges, namely δ , θ , α , β , and lower γ bands by considering the Welch's method (Welch, 1967). Hjorth features (Hjorth, 1970) are a set of time-domain features represented by the following three parameters: activity, mobility and complexity. Activity corresponds to the variance of the signal, indicating its overall magnitude. Mobility describes the mean frequency or the rate at which the signal changes, while complexity can be seen as the signal's irregularity, or the complexity of its waveform. Considering the 4 EEG acquisition channels from which the features are extracted, there are a total of 4×3 Hjorth features and 4×5 selected frequency bands for PSD computation, amounting to a final set of 32 EEG features.

4. Results

4.1. Questionnaires

Table 1 reports mean scores for each sound condition, as well as global means and medians. To test the statistical significance of these results, one-way repeated measures ANOVAs were conducted to assess the effect of sound condition on participants' subjective experiences.

Cronbach's α values indicated that the Competence scale showed acceptable internal reliability ($\alpha = 0.76$). Flow and Audio Immersion demonstrated borderline yet still adequate consistency ($\alpha = 0.67$ and $\alpha = 0.60$, respectively). In contrast, the Challenge scale showed low reliability ($\alpha = 0.46$), suggesting limited coherence of the construct in the context of this brief gaming task. These reliability levels should be taken into account when interpreting the following statistical findings.

A significant main effect of sound conditions was found on *audio immersion*, $F(2, 58) = 5.25$, $p = 0.008$, $\eta^2 = 0.108$, and *flow*, $F(1.72, 49.91) = 4.07$, $p = 0.022$, $\eta^2 = 0.044$, with borderline sphericity violation for the latter ($\epsilon = 0.81$, $p_{\text{spher}} = 0.022$). No significant main effects were found for the other dimensions.

Shapiro-Wilk tests indicated that *flow* violated normality for the inconsistent diegetic audio condition, which may explain why the corresponding Friedman test, performed as a sensitivity analysis, did not reach significance ($p = 0.106$), whereas the ANOVA remained significant. For all other dimensions, Friedman tests produced results consistent with the ANOVAs; these non-parametric results are reported in Table A2 in Appendix A.

Post-hoc power analyses based on the observed effect sizes indicated very high power for *competence* ($\eta_g^2 = 0.097$, $\text{power} = 0.987$) and *audio immersion* ($\eta_g^2 = 0.107$, $\text{power} = 0.995$), moderate power for *flow* ($\eta_g^2 = 0.044$, $\text{power} = 0.728$), and low power for *challenge* ($\eta_g^2 = 0.005$, $\text{power} = 0.115$), *liking* ($\eta_g^2 = 0.001$,

Table 1. Questionnaire components by sound condition (mean and standard deviation).

Component	Consistent sound	No sound	Inconsistent sound	Global mean	Global median
Competence	3.44 (0.73)	3.55 (0.75)	3.69 (0.73)	3.56 (0.76)	3.6
Challenge	3.28 (0.66)	3.17 (0.77)	3.21 (0.59)	3.22 (0.67)	3.25
Flow	3.39 (0.60)	3.12 (0.55)	3.37 (0.62)	3.29 (0.59)	3.29
Audio immersion	3.60 (0.74)	3.74 (0.64)	3.13 (0.88)	3.48 (0.80)	3.5
Liking	6.16 (1.82)	6.06 (1.79)	6.13 (1.78)	6.12 (1.78)	6.5
Valence	5.70 (1.89)	5.97 (1.56)	6.00 (1.78)	5.88 (1.73)	6
Arousal	4.66 (2.26)	4.93 (2.37)	5.16 (2.15)	4.92 (2.25)	6

Note that the last 3 components range 1–9, whereas the others range 1–5.

power = 0.068), *valence* ($\eta_g^2 = 0.009$, *power* = 0.180), and *arousal* ($\eta_g^2 = 0.008$, *power* = 0.175), suggesting that the significant effects for *audio immersion* are robust, the *flow* effect is moderately powered, and the non-significant effects for other dimensions may be due to limited statistical power.

Bonferroni-corrected pairwise *t*-tests were conducted to further examine significant ANOVA effects. Participants reported significantly greater *audio immersion* both in the consistent audio ($t(29) = 2.63$, $p_{\text{bonf}} = 0.041$, $d = 0.58$) and no-audio conditions ($t(29) = 2.86$, $p_{\text{bonf}} = 0.023$, $d = 0.79$) when compared to inconsistent audio game session, while no difference emerged between the consistent and no-audio conditions. Pairwise comparisons also indicated higher *flow* in the consistent audio condition compared to the sound-off one ($t(29) = 2.32$, $p_{\text{bonf}} = 0.082$, $d = 0.48$), and marginally lower flow in the latter vs. inconsistent condition ($t(29) = -2.08$, $p_{\text{bonf}} = 0.138$, $d = -0.42$), although none of them survived Bonferroni correction.

Despite no significant main effects were found for participants' self-reported *arousal* ($F(2, 58) = 2.20$, $p = 0.120$, $\eta_g^2 = 0.008$), for exploratory purposes we highlight that corrected pairwise comparison indicated significantly higher arousal during inconsistent sound game session compared to the consistent one ($t(29) = -2.81$, $p_{\text{bonf}} = 0.026$).

4.2. EEG signals

4.2.1. Effect of sound condition on EEG features across the four electrodes

A series of repeated-measures ANOVAs and *post-hoc* pairwise comparisons were conducted to examine condition-related effects on EEG features across individual channels (TP9, TP10, AF7, AF8). Specifically, EEG features included five PSD bands (δ , θ , α , β , and lower γ) and three Hjorth parameters (Activity, Mobility, and Complexity). To improve interpretability, repeated-measures ANOVAs results are reported in Table 2 only for features that resulted significant. For post-hoc comparisons, results are grouped by feature type (e.g., frequency bands, Hjorth parameters) and presented below based on the consistency of the observed effects across different EEG channels.

Broadly, for the α , β , and γ frequency bands, significant differences or relevant effect sizes could be detected on temporo-parietal channels (also at ANOVA level), whereas no significant results could be associated with frontal lobe activity. Particularly, on the γ band, TP10 showed a significant difference between the consistent audio and no-sound conditions ($p_{\text{corr}} = 0.003$, $d = 0.32$), and TP9 showed borderline significant differences for both consistent audio and no-sound, as well as consistent audio and inconsistent audio ($p_{\text{corr}} \approx 0.05$ – 0.06), with moderate effect sizes ($d \approx 0.33$ – 0.36). For the β band, significant effects were observed in both temporo-parietal channels across all condition pairs. In contrast, no difference remained statistically significant after multiple comparison correction for the α band,

Table 2. ANOVA results for each significant feature-channel pair.

Feature	Channel	F-value	p-Value	Generalized η^2
Delta	TP10	4.61	0.025*	0.105
	AF7	5.43	0.007	0.134
	AF8	7.53	0.003*	0.185
Theta	TP10	3.73	0.008	0.168
	AF7	3.82	0.036*	0.072
	AF8	3.31	0.055*	0.072
Alpha	TP9	3.53	0.036	0.055
	TP10	4.89	0.011	0.078
Beta	TP9	16.86	<0.001	0.049
	TP10	16.46	<0.001	0.055
Gamma	TP9	3.85	0.027	0.029
	TP10	4.59	0.014	0.019
Activity	TP9	5.87	<0.01*	0.167
	TP10	6.47	<0.01*	0.181
	AF7	4.43	0.016	0.131
	AF8	4.81	0.022*	0.141
Mobility	AF7	5.78	0.005	0.053
	AF8	7.16	0.001	0.865
Complexity	AF7	7.45	0.001	0.069
	AF8	7.04	0.002	0.055

Asterisk (*) indicates *p*-values that were corrected for violations of sphericity.

although moderate effect sizes were observed in the comparison between consistent audio and no-sound at TP9 ($d = 0.46$, $p = 0.113$) and TP10 ($d = 0.53$, $p = 0.068$). Additionally, a similar effect size was observed in TP10 between the two audio-on sessions ($d = 0.55$, $p = 0.111$), indicating potential trends worth further investigation.

In contrast to these frequency bands, Mobility and Complexity showed significant results in the frontal channels (also at the ANOVA level), with no significant effects registered in the temporo-parietal area. For both features, a significant difference was found on AF7 between the consistent audio and no-sound conditions (both $p = 0.002$). In AF8, significance was found for Mobility not only between the consistent audio and no-sound conditions, but also between consistent audio and inconsistent audio ($p < 0.001$, and $p = 0.025$, respectively). The same contrasts were significant for AF8 in Complexity, with differences between consistent audio and no-sound ($p = 0.001$), and between consistent and inconsistent audio ($p = 0.047$).

Results are more heterogeneous for the remaining features. Significant differences were observed in Activity across all channels at ANOVA level, and for one or two pairs of conditions through *post-hoc* comparisons. Particularly, the comparison between consistent audio and no-sound was significant across all channels (AF7: $p = 0.011$; AF8: $p = 0.013$; TP9: $p = 0.023$; TP10: $p = 0.025$), with TP10 also showing significance between the two sound-on conditions ($p = 0.048$). In the δ band, strong condition effects were found between consistent audio and no-sound in TP10 ($p_{\text{corr}} = 0.043$, $d = 0.75$), as well as in both frontal channels, which showed large effect sizes (AF7: $p_{\text{corr}} = 0.0023$, $d = 0.96$; AF8: $p_{\text{corr}} = 0.0016$, $d = 1.11$). These channels were also significant at ANOVA level. Finally, in the θ band, the contrast between consistent audio and no-sound was significant at AF7, ($p_{\text{corr}} = 0.009$, $d = 0.70$), while a trending difference was observed at TP9, ($p_{\text{corr}} = 0.114$, $d = 0.49$), indicating a moderate effect that did not reach significance after correction. Significance in these channels differs from that resulting in TP10, AF7, and AF8 at the ANOVA level.

4.2.2. Effect of sound condition on EEG features aggregated by channel

A repeated-measures ANOVA was conducted to assess the effect of the sound conditions also on the EEG features averaged across the four acquisition channels. As depicted in Table 3, a significant main effect of condition was found for all the features with the exception of PSD on β and γ waves.

Post hoc tests (Bonferroni-corrected) revealed that, despite a main effect of the audio stimuli in the α power, $F(2, 58) = 3.85$, $p = 0.027$, $\eta_G^2 = 0.07$, no pairwise comparisons survived Bonferroni correction. On the other hand, there was a significant difference between the consistent audio and sound-off conditions for δ ($p_{\text{bonf}} = 0.001$, $d = 1.11$) and θ ($p_{\text{bonf}} = 0.017$, $d = 0.72$) powers, as well as for all the Hjorth features (Activity: $p_{\text{bonf}} = 0.004$, $d = 1.14$; Mobility: $p_{\text{bonf}} = 0.044$, $d = -0.28$; Complexity: $p_{\text{bonf}} = 0.003$, $d = 0.53$). Moreover, although no significant main effect of sound conditions was found for the γ band ($F = 2.61$, $p = 0.082$), also in this case contrast between these two conditions reached significance ($p_{\text{bonf}} = 0.032$, $d = 0.29$). No other contrasts resulted significant for the discussed features.

Interestingly, the polarity of the *T*-statistics exhibited a consistent pattern across features: it was negative for the contrast between the no-audio and inconsistent-audio conditions, and positive for the contrasts involving the congruent-audio condition versus no audio or inconsistent audio. The only exception is Mobility, which exhibits the opposite trend.

Table 3. ANOVA results for EEG features across conditions.

Feature	<i>F</i> -value	<i>p</i> -Uncorrected	Generalized η^2
Delta	7.79	0.0010	0.1828
Theta	4.90	0.0108	0.0917
Alpha	3.85	0.0269	0.0677
Beta	1.21	0.3069	0.0158
Gamma	2.61	0.0821	0.0162
Activity	7.00	0.0019	0.1930
Mobility	3.45	0.0383	0.0137
Complexity	6.80	0.0022	0.0430

Significant *p*-values are reported in bold. Note that for Delta power the sphericity condition is violated ($\epsilon = 0.82$, $p_{\text{spher}} = 0.033$), and the corrected *p*-value remains significant, amounting to 0.0022.

Table 4. Performance metrics for Audio Immersion and Flow classification.

Phase	Feature set	Audio immersion			Flow		
		Acc	Prec	F1	Acc	Prec	F1
Train	Full	0.7437	0.6378	0.6614	0.7823	0.7119	0.7161
	Reduced	0.8988	0.8489	0.8659	0.8659	0.8494	0.8354
Test	Full	0.4222	0.3463	0.3488	0.5555	0.5555	0.5266
	Reduced	0.5444	0.5370	0.5078	0.5777	0.6278	0.5611

4.3. XGBoost and features importance

As reported in Section 4.1, analysis of the questionnaire data revealed that both *flow* and *audio immersion* had global mean and median scores of approximately 3.5. This value was therefore selected as a threshold for binarising the self-reported responses into high (Likert scale scores of 4 or 5) and low (scores of 1, 2, or 3) conditions for both *flow* and *audio immersion*. After binarisation, the resulting classes were relatively balanced, with 51.1% of samples classified as high in *audio immersion* and 40% as high in *flow*.

Performance results for both the training and test sets, using both the full and reduced feature sets, are summarized in Table 4. For each dimension, feature selection retained only those features with a gain score above the mean gain score, resulting in a subset of 15 elements for *audio immersion* ($M = 2.60$, $SD = 0.44$) and 13 elements for *flow* ($M = 2.74$, $SD = 0.58$). Regardless of the utilized features set, performance suffered from the small sample size and the chosen splitting validation strategy. Using the full feature sets, training accuracy ranged between 63.78% and 78.23%, while test performance remained close to chance level, being slightly above 0.5 for *flow* and below 0.5 for *audio immersion*. In contrast, feature reduction yielded a consistent improvement in performance across all metrics in both training and test sets. Particularly, accuracy on the training set reached 89.88% for *audio immersion* and 86.59% for *flow*, with corresponding test accuracies of 54.44% and 57.77%, respectively.

Since the primary objective of this analysis is to identify EEG features that could be influential in separating high/low levels of immersion and flow, the focus is placed on feature importance analysis rather than model performance. Figure 3 displays the gain scores of the retained features in descending order, whereas Tables 5 and 6 report the number of retained features by feature type and channel location for *audio immersion* and *flow*, respectively.

A noteworthy finding concerns the contribution of the different types of features, with PSD features being the great majority of retained features for *flow* (77%), whereas Hjorth parameters represent only slightly more than half (53.3%) of retained features for *audio immersion*. Interestingly, although fewer Hjorth features were retained for *flow*, those included tend to exhibit higher gain scores than their PSD counterparts.

For *audio immersion*, in terms of spatial distribution, Hjorth features were relatively evenly spread across cortical areas (frontal and temporal lobes), whereas 5 out of the 7 retained PSD features were localized in the frontal region. The most informative features were: Activity at AF8, and Mobility and Complexity at AF7, all of which were previously found significant between consistent and sound-off conditions, as well as β at AF7 and Complexity at TP10, which in contrast did not show significance in prior statistical tests.

In the selected features for *flow*, there appears to be hemispheric lateralization with 9 out of the 13 retained features originating from the left hemisphere. A standout result was the dominance of θ power at AF7, with a gain score exceeding 4, which is well above any other retained feature. The next most important features were β at TP10, which was also significant across all condition pairs, and γ at AF7, which did not reach significance in statistical tests.

5. Discussion

5.1. Questionnaires

Based on questionnaire results, participants felt immersed into the game not only, as expected, while both diegetic and extradiegetic audio were present and consistent with gameplay, but also when sound was completely turned off. In fact, as noted in comments collected after the experiment, some of the

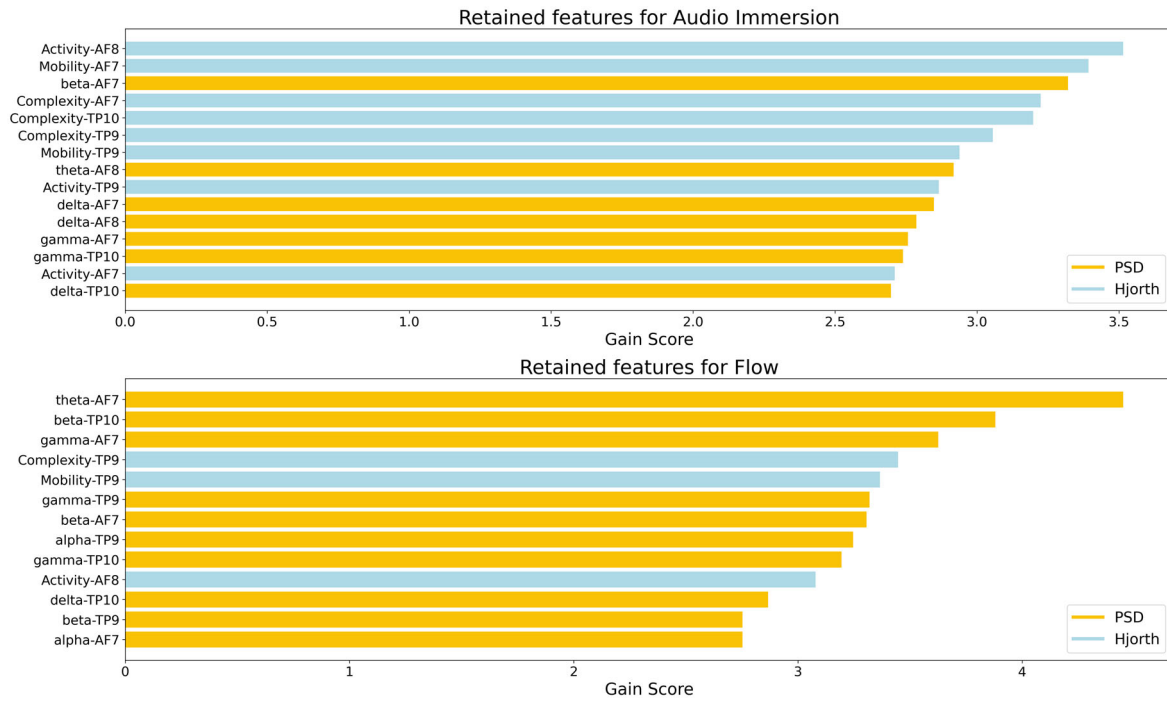


Figure 3. Retained features above mean gain score for each target dimension.

Table 5. Count of EEG features occurrences by channel and feature type for Audio Immersion.

Channel	PSD						Hjorth parameters			
	Delta	Theta	Alpha	Beta	Gamma	Tot.	Activity	Mobility	Complexity	Tot.
TP9	0	0	0	0	0	0	1	1	1	3
TP10	1	0	0	0	1	2	0	0	1	1
AF7	1	0	0	1	1	3	1	1	1	3
AF8	1	1	0	0	0	2	1	0	0	1
Tot.	3	1	0	1	2	7	3	2	3	8

Table 6. Count of EEG features occurrences by channel and feature type for Flow.

Channel	PSD						Hjorth parameters			
	Delta	Theta	Alpha	Beta	Gamma	Tot.	Activity	Mobility	Complexity	Tot.
TP9	0	0	1	1	1	3	0	1	1	2
TP10	1	0	0	1	1	3	0	0	0	0
AF7	0	1	1	1	1	4	0	0	0	0
AF8	0	0	0	0	0	0	1	0	0	1
Tot.	1	1	2	3	3	10	1	1	1	3

participants preferred or were accustomed to playing video games without audio, and therefore declared an increased immersion due to sound absence. This result highlighted the need to consider this habit as a criterion for participants selection in future studies, to include a uniform pool of subjects from this point of view. The significantly reduced immersion during mismatched diegetic audio compared to the other conditions is particularly relevant in that it suggests that the physiological differences observed between the two sound-on conditions may be attributed to the differing self-reported immersion levels. However, the same result should be considered with caution when comparing inconsistent and sound-off sessions, as EEG data analysis should account for the presence or absence of audio. In fact, this may influence neural signals, especially in temporal regions such as those measured by temporo-parietal electrodes. It is worth noting that the highest mean Audio Immersion score ($M = 3.74$, $SD = 0.64$) was associated with the sound-off condition. This finding contrasts with previous literature, such as (Grimshaw et al., 2008), which reported that the complete absence of sound tends to significantly reduce the subjective sense of immersion. This discrepancy may be attributed to differences in game design and genre, considering that their study was based on a FPS game.

Regarding flow, subjective responses indicated that neither the presence nor the consistency of the audio influence it in a marked way, suggesting that entering this state might be independent of the tested sound conditions. However, the presence of coherent sound might still enhance the degree of flow, even if this was not statistically significant in our results after correction, which is possibly due to the short duration of the game sessions. We acknowledge that longer gameplay sessions may provide a more sensitive framework for eliciting sustained flow states and amplifying condition-dependent differences, particularly in relation to audio coherence. The present findings therefore highlight session duration as a potentially important factor to consider when designing future studies aimed at capturing sound-related effects on flow. The fact that flow could be experienced even in the presence of inconsistent audio could be attributed to (i) habituation, as players adapt to the unexpected sound associations and begin to interpret them as consistent cues over time, and (ii) increased arousal. In fact, self-reported increased arousal in the inconsistent audio session compared to the normal one is coherent with some literature results (Gemba & Sasaki, 1988) and conclusions (Kenwright, 2020), which highlight the role of active sound in gameplay as a driver of heightened sensory arousal compared to more passive auditory experiences. While an increased arousal may also be expected in the consistent sound-on compared to the sound-off condition, the observed higher mean arousal in the latter with respect to the consistent sound session might suggest that, in the absence of sound, players were more susceptible to arousal elicited by in-game events alone.

Finally, regarding competence and challenge, the main concern was verifying that they did not significantly vary across experimental conditions considering all the subjects jointly, which was indeed guaranteed by the adopted subject-independent strategy and the randomized order of sessions administration across participants. In fact, as the manipulation of the game difficulty level impacts the balance between perceived challenge and perceived skills, which is claimed to be at the foundations of flow, difficulty level was not changed as the game proceeded in order not to confound the effect of sound.

5.2. EEG data

EEG analysis revealed a robust physiological distinction between the consistent diegetic audio and sound-off conditions. This was evident not only in the aggregated features but also in numerous combinations of EEG features and electrode sites: for all the high-frequency bands (α , β , and γ), no effects were found in frontal sites, while statistically significant differences emerged between these two conditions in the temporal lobe (moderate effect sizes only in the case of the α band); θ power was significantly different in the left anterior frontal area only, whereas δ power showed consistent significance across all electrodes. Regarding Hjorth features, both Mobility and Complexity were significantly different in frontal areas, whereas Activity, just like δ power, was significantly different across all electrodes.

Despite these clear physiological distinctions between consistent sound and sound-off conditions, no significant differences were reflected in the self-assessed game experience. This highlights the complexity of self-reporting subjective states such as immersion and flow, and the general limitation in questionnaire-based tools in capturing mental states whose theoretical definition in the literature not only lacks agreement, but often does not even clearly separate between them.

When the focus shifts to the comparison between consistent and inconsistent diegetic sound game sessions, i.e., the only pair of sessions for which subjective questionnaires reported a significant difference in the perceived immersion, fewer physiological features were found to be relevant. Particularly, these were a subset of those already significant between the previously discussed consistent sound versus sound-off conditions: β power in both temporal electrodes, Mobility and Complexity in the right frontal electrode, and Activity in the right temporo-parietal area, where a moderate α effect size was also detected. Notably, β power in the temporal lobe consistently differentiated all three pairs of conditions, with the highest values in the consistent sound condition and the lowest in the no-sound condition, possibly indexing audio-related processing more than immersion itself.

Beyond condition-specific findings, clear topographical and spectral patterns emerged. For higher frequency bands (α to γ), significant differences were observed in the temporo-parietal area, whereas for lower frequency bands (θ and δ), significance was found in the frontal and right temporo-parietal

lobes. For Hjorth parameters, Activity was significant across all electrodes, while Mobility and Complexity were significant only on the frontal ones.

These spatial associations of time and frequency features partially align with feature importance analysis: temporo-parietal high-frequency dynamics were decisive in characterizing the presence of Flow, while frontal low-frequency Activity and Hjorth features seemed more involved in discriminating Audio Immersion levels. However, 5 out of the 15 retained Audio Immersion features and 5 out of the 13 retained Flow features were not significant at ANOVA level, indicating that machine learning models may identify complementary information to that brought by statistical analyses. Despite classification performances on the test sets not being adequate in this case, the importance of implementing a feature analysis approach relies on the fact that it helps identify physiological biomarkers of the player's mental states independently of the specific auditory stimuli, guiding the distinction between high versus low levels of immersion and flow. This is particularly relevant considering that the experimental conditions did not induce a marked self-reported difference in the psychological states under study. The high gain score achieved by the θ power at AF7 in discriminating Flow, despite its lack of significance across the different audio sessions (except for the consistent versus sound-off ones), suggests that this feature, registered at this location, may be a biomarker for this mental condition. The numerous statistical significances observed mainly between consistent sound and sound-off conditions only further support the adoption of the features importance approach. In fact, these physiological differences suggest that the observed neural changes may not be attributed solely to the presence/absence of audio (as significant differences would have consequently been expected also between the inconsistent and sound-off conditions). Instead, they hint that differences may be indicative of a mental state that is modulated by both presence and congruence of sound and that is not adequately captured by self-reported experience.

Previous literature partially confirms the observed patterns in brain waves power analysis. Increased δ power has been associated with immersion (Nacke et al., 2011) and trance-like states (Boutros et al., 2008) and, in our study, δ was consistently and significantly higher in the congruent sound sessions rather than in the sound-off sessions across all channels. Increases in frontal θ activity mirror findings in affective ludology that link this band to immersion (Lim et al., 2019; Nacke et al., 2011), and to engagement and engrossment specifically in frontal areas (Kannegieser & Ratz, 2021). Particularly, the highest power in our case is detected in case of consistent diegetic audio, while the lowest for the sound-off condition, potentially suggesting decreased immersion and enjoyment under this game setting. The α band, often associated with waking rest and moderate concentration (Ray & Cole, 1985), did not reach significance but showed moderate effects in the right temporo-parietal area, potentially reflecting modulations in attentional engagement. Prior work (Kannegieser & Ratz, 2021) also reports that α and γ activity were not found to be relevant to immersion or flow, although in our study γ was significantly higher in the consistent diegetic audio condition than in the sound-off condition. β rhythms were found to be significantly higher during immersion compared to a boredom (Nacke et al., 2011), baseline (Kannegieser & Ratz, 2021), or, in the temporal lobes, a concentration condition (Lim et al., 2019). In our study, β powers were significantly different across all three conditions in the temporal areas, and followed a graded pattern (consistent diegetic audio > inconsistent diegetic audio > sound-off), which diverged from the participants' self-reported immersion ordering (sound-off > consistent diegetic audio > inconsistent diegetic audio). As β oscillations in the auditory cortex have been shown to reflect predictive timing (Arnal & Giraud, 2012; Fujioka et al., 2015) and also predictive coding of which sound stimulus is expected next (Chang et al., 2018), differences in β power observed in this study may indeed reflect, rather than immersion per se, auditory processing demands or attentional modulation due to sound structure. Our observation of temporal but no frontal differences aligns also with studies reporting that an absence of change in frontal β activity has been connected to measuring engaged enjoyment during different types of activities (Hurless et al., 2013; Lee & Hsieh, 2014). This further supports the interpretation that observed differences in the temporal lobe in this frequency band are to be attributed to the auditory stimuli and must therefore be regarded as such in future studies that will account for additional game elements. Finally, it is worth noticing that music that is personally liked by an individual is reported (Kučikienė & Praninskienė, 2018) to enhance EEG power

spectra globally and across bandwidths, and that such an effect is seen particularly in β frequencies of temporal regions. This finding reinforces our observed F-values, always highest in the consistent sound session and lowest when the sound was off, across all bandwidths, and with this trend being significantly different in the very β waves of temporal lobes. No previous work has assessed immersion or flow considering Hjorth parameters, marking an important contribution of this study. A remarkable finding concerns the pattern observed in Mobility, which is opposite compared to that of Activity, Complexity, and all spectral power bands. In fact, values were highest when audio was off and lowest during consistent audio, across all electrodes. This trend will be the subject of further investigation.

Overall, these results suggest that while many EEG features can discriminate between some of the experimental conditions (mainly between the presence of consistent diegetic audio and the complete absence of sound), associating such changes directly to immersion or flow is not straightforward as, in this case, some differences may likely arise from sensory or attentional mechanisms unrelated to subjective absorption. To address this ambiguity, experimental states need to be rethought to induce a more separable subjective response in the questionnaires, to be possibly correlated to the corresponding registered physiological responses with a major confidence. At the same time, the difficulty in mapping physiological patterns and self-reported experiences raises questions about the adequacy of current ground truth measures. In fact, given the inherently introspective nature of constructs like immersion and flow, as well as the lack of clear distinction in their conceptualisations, their quantification through questionnaires may fail to capture their full complexity. This highlights the need for refined questionnaires with a clear definitional premise, to ultimately develop a real-time detection system which may obviate the need for questionnaire-based assessments.

6. Conclusions

This study analyzed immersion and flow in video games in response to different auditory stimuli by combining self-report questionnaires with EEG signal analysis. The underlying premise was that no reliable physiological marker has yet been established in the literature to objectively measure these complex states. Given the multifactorial nature of flow and immersion in relation to the video game experience, a modular approach was adopted to isolate a single component of gameplay at the time (in this case, sound) to control for it. The idea was that such an approach could help disentangle the specific impact on neural responses of each video game immersive component. A detailed analysis of EEG responses across conditions was performed, along with an exploration of their correspondence with self-reported questionnaire outcomes.

At the physiological level, differences among sound conditions emerged as relatively limited in scope (both in terms of their frequency and time-frequency characteristics) and spatially localized at the cortical level. A robust difference in physiological response was found between the consistent sound and sound-off conditions. However, this neural separation did not clearly map onto the participants' self-reported experiences, indicating a possible disconnect between subjective recall and physiological data. Consequently, future analyses will, on one hand, focus on the specific EEG features and electrode sites that aligned with conditions found to differ significantly at both physiological and subjective levels. On the other hand, it will be important to investigate whether the lack of correspondence in certain cases reflects limitations in the sensitivity of the questionnaires, or whether participants struggled to accurately introspect their immersive experience, and more marked differences among experimental conditions would be needed. Moreover, individual differences in game-related habits, particularly familiarity with the game's genre and the tendency to play without audio, may act as moderators influencing both subjective and physiological responses. Building on this point, as individual preferences in gameplay habits (such as being accustomed to playing without sound) may influence both subjective and physiological responses, and as genre familiarity may reduce cognitive load or shape expectations, stratification of participants by auditory preference or familiarity with both the genre and audio-use habits could enhance homogeneity within same experimental condition, and potentially increase separation between different ones. Although the present study did not stratify participants based on these factors, acknowledging their potential moderating role is important, and future work should incorporate stratified or moderator analyses to better isolate sound-related effects. Additionally, while the current

validation strategy was designed to generalize findings across individuals, rather than to identify individual-specific patterns, future applications aimed at real-time adaptation will require a more personalized perspective. Given the highly subjective nature of immersive experiences and the individual variability in physiological signals, subject-specific analyses may reveal stronger, more nuanced differences and enhance the feasibility of personalized biofeedback systems.

We acknowledge that these are preliminary findings, yet they represent an important step toward identifying objective markers of game-related immersion and flow in ecologically valid contexts, and offer a foundation for the future development of adaptive game systems that dynamically respond to the player's state in real time.

Notes

1. Godot Engine official website: godotengine.org.
2. Nintendo eShop product page (Arcade Archives: Mario Bros.): nintendo.com/us/store/products/arcade-archives-mario-bros-switch/.
3. Shimmer Sensing official website: shimmersensing.com.
4. Muse Headband official website: choosemuse.com.
5. MathWorks MATLAB product page: mathworks.com.
6. EEGLAB toolbox (MATLAB File Exchange): mathworks.com/matlabcentral/fileexchange/56415-eeqlab.

Author contributions

CRedit: **Sara Nocco**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Marco Cremaschi**: Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was partially supported by the MUR under the grant “Dipartimenti di Eccellenza 2023-2027” of the Department of Informatics, Systems and Communication of the University of Milano-Bicocca, Italy.

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Appendix A. Post-game questionnaire

Table A1. Post-game questionnaire.

#	Item
1	How do you rate the difficulty of the game?
2	How much did you want to win the game?
3	Do you think the game was challenging?
4	Did you feel motivated while playing?
5	Did you feel in full control of your movements in the game?
6	How much effort did you put into playing?
7	Did you feel you were doing your best?
8	How do you evaluate your performance in the game?
9	To what extent did you feel like you were making progress toward completing the game?
10	To what extent were you distracted by external sounds?
11	Did you feel the need at any point to stop playing in order to see or hear what was happening around you?
12	To what extent have you enjoyed the game audio?
13	To what extent have you felt bothered by the game audio?
14	To what extent have you found the audio entertaining?
15	To what extent did the audio help you feel immersed in the game?
16	To what extent did the audio distract you during gameplay?
17	To what extent did you adapt to the audio as the game progressed?
18	To what extent did the game keep your attention?
19	How aware were you of yourself in the space around you?
20	Did you lose track of time while playing?
21	How involved and focused did you feel while playing?
22	To what extent were you aware of being in the real world while playing?
23	To what extent was your feeling of being inside the game environment stronger than the feeling of being in the real world?
24	Did you notice events happening around you?
25	Valence: Did you feel happy?
26	Arousal: Did you feel excited?
27	Liking: How much did you like the game overall?

Table A2. Friedman test results for questionnaire components across sound conditions.

Component	W	Q	p-Value
Mean challenge	0.01186	0.71154	0.70063
Mean competence	0.33856	20.31373	<0.001
Mean audio immersion	0.11780	7.06780	0.02919
Mean flow	0.07462	4.47706	0.10662
Liking	0.00132	0.07895	0.96130
Valence	0.02149	1.28947	0.52480
Arousal	0.06558	3.93478	0.13982